

Penn Aerial Robotics - Plane fit

- 1) a) ~~current flows from high voltage to low voltage. A voltage differential causes~~

Let $z = a + bx + cy$

$$E(a, b, c) = \sum_{i=1}^n (z_i - a - bx_i - cy_i)^2$$

$$\frac{\partial E}{\partial a} = -2 \sum_{i=1}^n (z_i - a - bx_i - cy_i) = 0$$

zero, since symmetric window

$$\sum_{i=1}^n z_i - an - \sum_{i=1}^n bx_i - \sum_{i=1}^n cy_i = 0$$

$$\sum_{i=1}^n z_i - an = 0$$

$$a = \frac{\sum_{i=1}^n z_i}{n}$$

$$\frac{\partial E}{\partial b} = -2 \sum_{i=1}^n x_i (z_i - a - bx_i - cy_i) = 0$$

$$-\sum_{i=1}^n x_i z_i + \sum_{i=1}^n ax_i + \sum_{i=1}^n bx_i^2 + \sum_{i=1}^n cx_i y_i = 0$$

$$b \sum_{i=1}^n x_i^2 = \sum_{i=1}^n x_i z_i$$

$$b = \frac{\sum_{i=1}^n x_i z_i}{\sum_{i=1}^n x_i^2}$$

$$\frac{\partial E}{\partial c}$$

$\approx$

$$c = \frac{\sum_{i=1}^n y_i z_i}{\sum_{i=1}^n y_i^2}$$

$$E_{\text{max}} = \sum_{i=1}^n \left(z_i - \frac{\sum_{i=1}^n z_i}{n} - \frac{\sum_{i=1}^n x_i z_i}{\sum_{i=1}^n x_i^2} x_i - \frac{\sum_{i=1}^n y_i z_i}{\sum_{i=1}^n y_i^2} y_i \right)^2$$

← This procedure is done parallelly but Independently for each pixel!

$$P_i = a + bx_i + cy_i$$

$$E = \sum (z_i - P_i)^2 = \sum z_i^2 - 2 \sum z_i P_i + \sum P_i^2$$

$$\sum z_i (a + bx_i + cy_i)$$

$$a \sum z_i + b \sum x_i z_i + c \sum y_i z_i$$

Known values

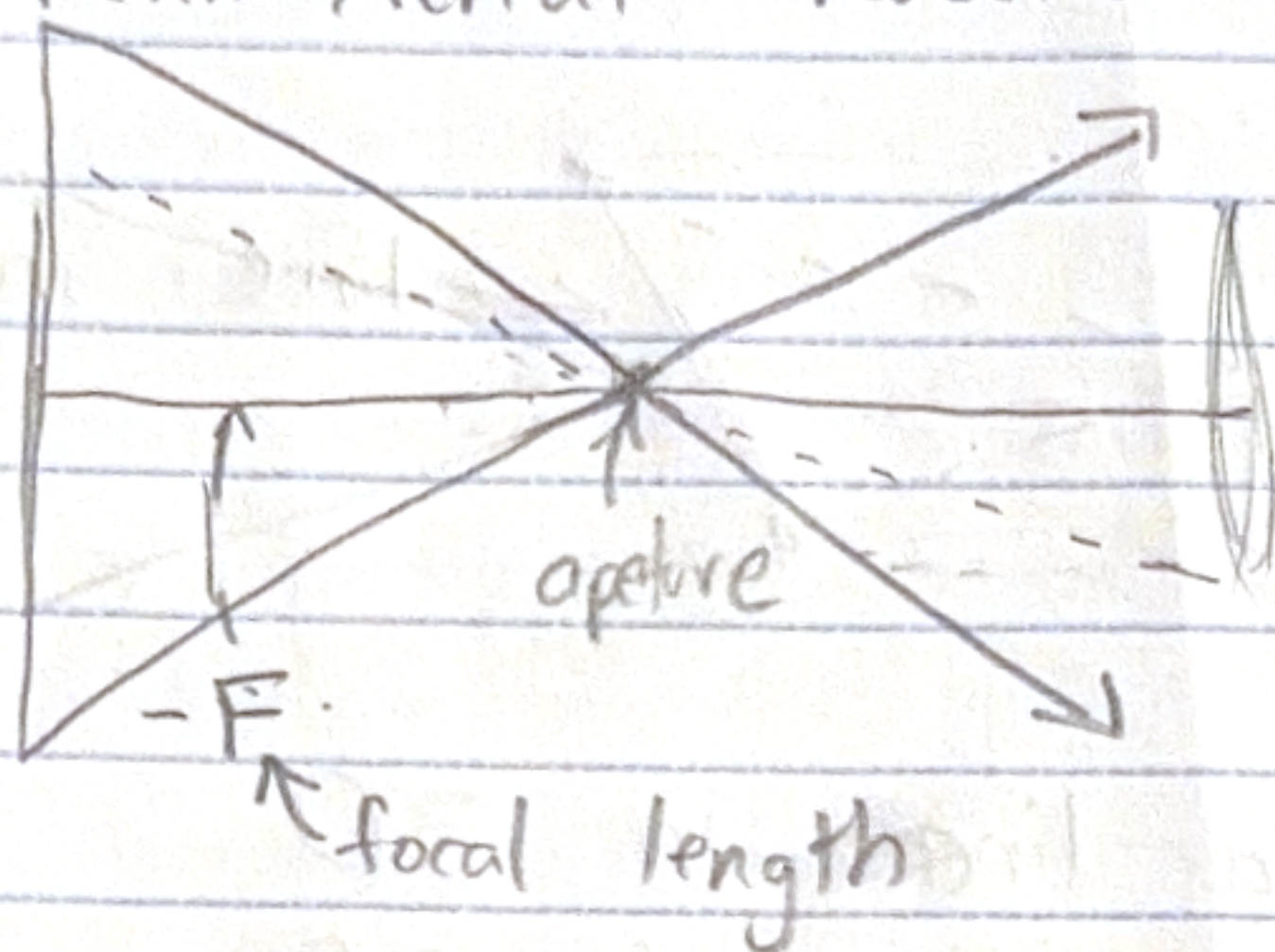
$$\sum P_i^2 = \sum (a + bx_i + cy_i)^2 = \sum a^2 + 2ab \sum x_i + 2ac \sum y_i + 2bc \sum x_i y_i + \sum b^2 x_i^2 + \sum c^2 y_i^2$$

$$= a^2 n + b^2 \sum x_i^2 + c^2 \sum y_i^2$$

$$= a \sum z_i + b \sum x_i z_i + c \sum y_i z_i$$

$$E = \sum z_i^2 - a \sum z_i - b \sum x_i z_i - c \sum y_i z_i$$

Penn Aerial Robotics : Camera Matrix



$$K = \begin{pmatrix} f_x & s & x_0 \\ 0 & f_y & y_0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$F_x = f_x = \frac{W}{w}, \quad W = \text{screen (mm)} \\ w = \text{pixels}$$

$$K \times X_0 = X_i; \quad K = \text{matrix (intrinsic)} \quad X_0 = \text{world coords (in)} \\ X_i = \text{pixels (px)}$$

$$\text{given } K = \begin{pmatrix} 2564 & 0 & 960 \\ 0 & 2570 & 540 \\ 0 & 0 & 1 \end{pmatrix} \rightarrow \text{no shear} \quad \Delta x = 1920 \\ \Delta y = 1080$$

$$f_x = 2564$$

$$f_y = 2570$$

$$W = 5 \text{ in} : w = 104.21 \text{ px} \leftarrow \text{circle radius}$$

$$F_x = \frac{2564}{104.21} \times 10 = 24.604 \text{ in} \quad F_y = \frac{2570}{104.21} \times 10 = 24.662 \text{ in}$$

$$\rightarrow X_0 = K^{-1} X_i = \begin{pmatrix} 1/2564 & 0 & -240/641 \\ 0 & 1/2570 & -54/257 \\ 0 & 0 & 1 \end{pmatrix} \times \begin{pmatrix} Sx_i \\ Sy_i \\ S \end{pmatrix} = \begin{pmatrix} \frac{Sx_i}{2564} - \frac{240S}{641} \\ \frac{Sy_i}{2570} - \frac{54S}{257} \\ S \end{pmatrix}$$

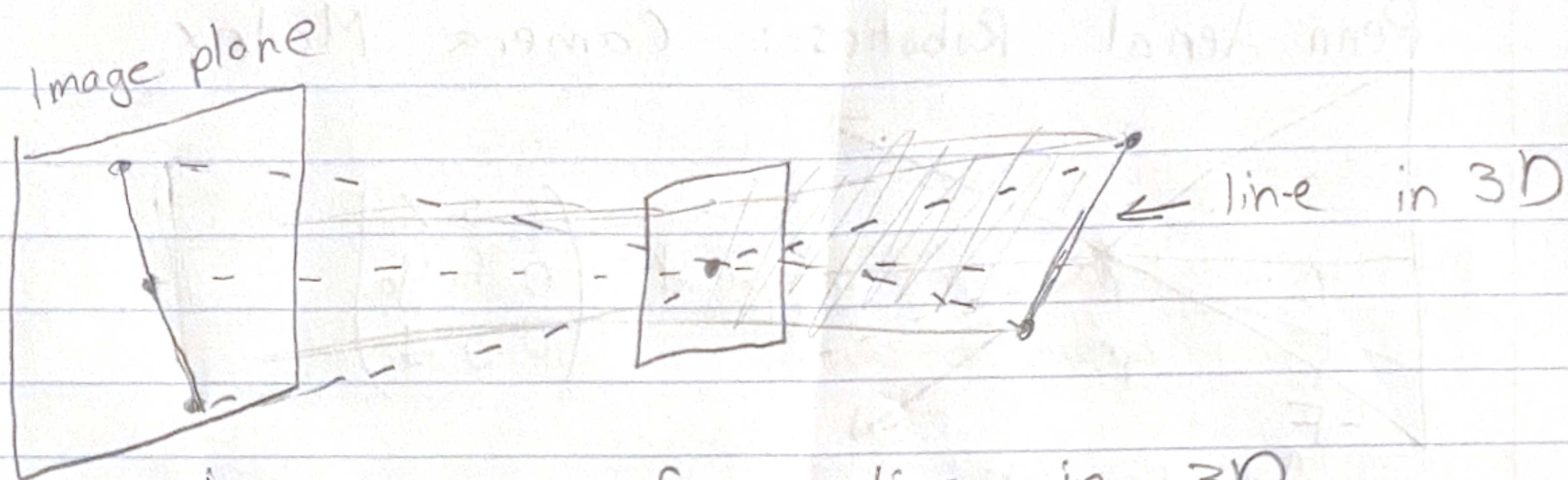
$$\rightarrow X_0 = S \begin{pmatrix} (x_i - 960)/2564 \\ (y_i - 540)/2570 \\ 1 \end{pmatrix}$$

solve for S:

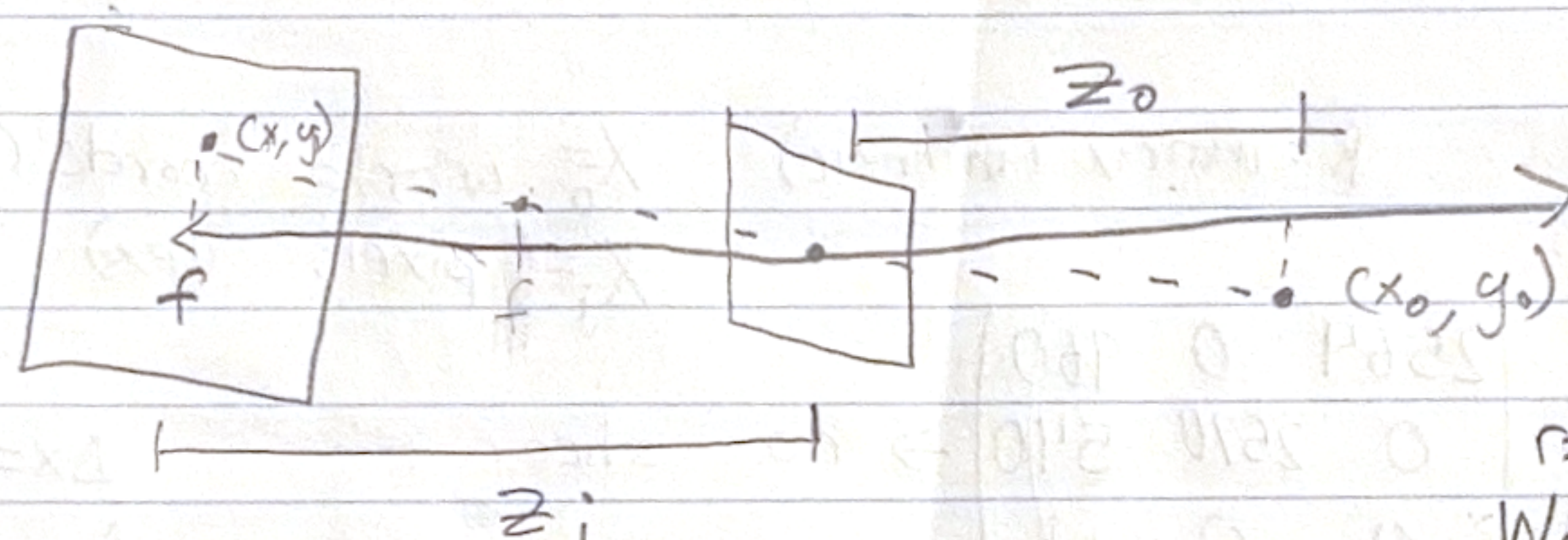
$$\text{when } (x_i, y_i) = (960 + 105, 540) \\ (x_0, y_0) = (5, 0)$$

$$5 = \frac{105S}{2564} \quad S \approx 122$$

$$\therefore \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} = \begin{pmatrix} 122 (x_i - 960)/2564 \\ 122 (y_i - 540)/2570 \end{pmatrix}$$



• An image of a line in 3D is same as a line in 2D.



$$\frac{f_x}{x_i} = \frac{z_0}{x_o} \quad \frac{f_y}{y_i} = \frac{z_0}{y_o}$$

$$x_i = 104 \text{ px} \quad x_o = 5 \text{ in} \\ f_x = 2564 \text{ px} \quad z_o = ?$$

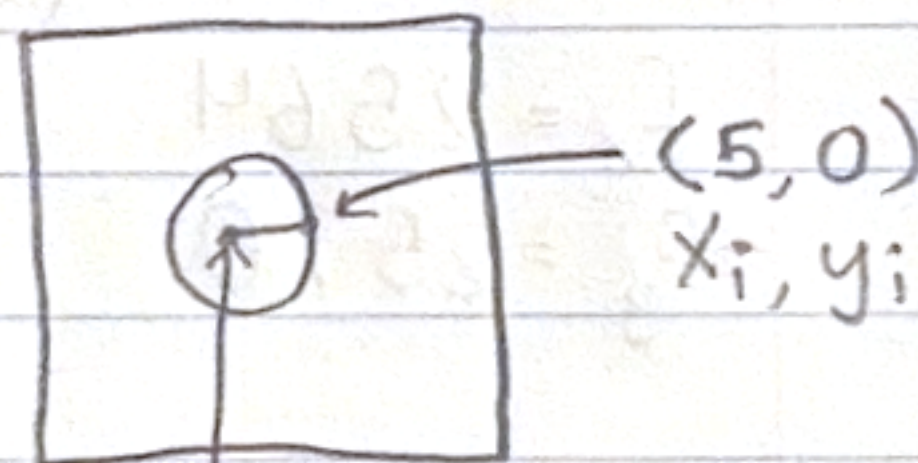
$$z_o = \frac{x_o}{x_i} f = \frac{(5 \text{ in}) (2564 \text{ px})}{(104 \text{ px})}$$

$$z_o \approx 123 \text{ in.} \leftarrow \text{solve for } z_o$$

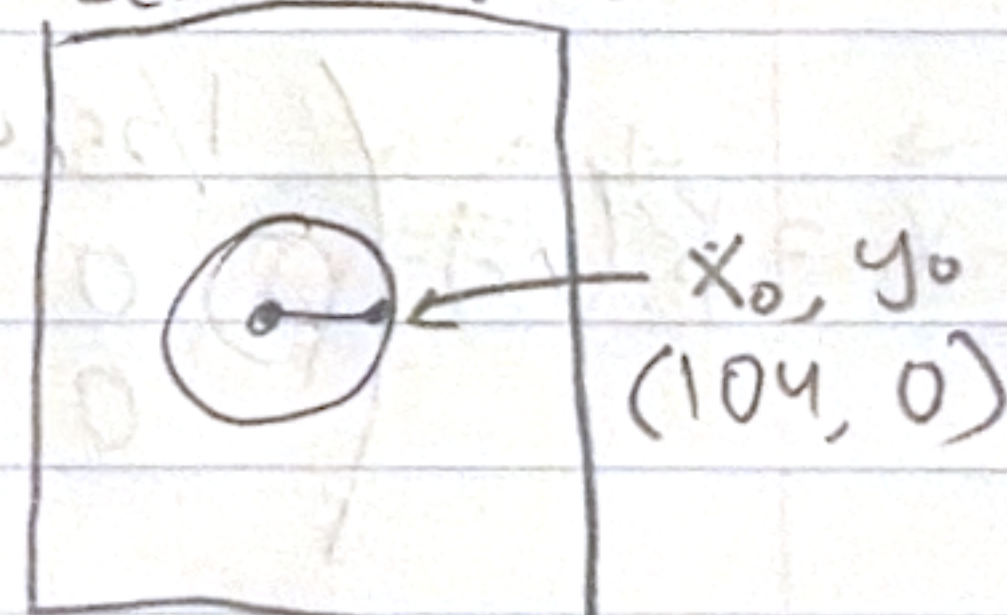
$$x_o = \frac{z_o x_i}{f_x} = \frac{123}{2564} (x_i - 960)$$

$$y_o = \frac{z_o y_i}{f_y} = \frac{123}{2570} (y_i - 540)$$

$r = 5 \text{ in}$
World Space



Screen Space



$r = 104 \text{ px}$

recall we shifted by $\langle 960, 540 \rangle$